

Graph Neural Diffusion

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Prerequisites

Message passing neural networks

One round of message passing for node v :

$$x_v^{k+1} = \phi\left(x_v^k, \bigoplus_{u \in \mathcal{N}(v)} \psi(x_v^k, x_u^k)\right),$$

where ϕ, ψ are shared «update» and «message» functions and $\bigoplus$ is any commutative

Diffusion equation

$$\frac{\partial x(u, t)}{\partial t} = \operatorname{div}[a \cdot \nabla x(u, t)]$$

Differential operators on graphs.

Let $X \in \mathbb{R}^n$ be node features.

► **Gradient:**

$$(\nabla X)_{ij} = x_j - x_i$$

► **Divergence:**

$$(\operatorname{div} F)_i = \sum_{j:(i,j) \in E} f_{ij}$$

► **Laplacian:**

$$(\Delta X)_i = (\operatorname{div}(\nabla X))_i = \sum_{j:(i,j) \in E} (x_j - x_i)$$

Diffusion Equation of Graphs

Node wise form

$$\frac{\partial}{\partial t} x_i(t) = \sum_{j:(i,j) \in E} a(x_i(t), x_j(t)) (x_j(t) - x_i(t))$$

Matrix form

$$\frac{\partial}{\partial t} X(t) = (A(X(t)) - I)X(t) = \bar{A}(X(t))X(t)$$

Forward euler discretisation (Explicit)

$$\frac{X^{(k+1)} - X^{(k)}}{\tau} = \bar{A}(X^{(k)})X^{(k)} \implies X^{(k+1)} = (I + \tau \bar{A}(X^{(k)}))X^{(k)}$$

Transition matrix has the same sparse structure as the adjacency matrix.

Backward euler discretisation (Semi-Implicit)

$$B^{(k)}X^{(k+1)} = X^{(k)}$$

Requires approximately inverting B thus resulting matrix is dense

Equivalence to GNNs

The forward Euler step is structurally identical to a GNN layer with a residual connection

$$x_i^{(k+1)} = x_i^{(k)} + f \left(\sum_{j:(i,j) \in E} g(x_i^{(k)}, x_j^{(k)}) \right)$$

- ▶ $\bar{A}(t) = \bar{A} \implies$ graph convolution network
- ▶ $\bar{A}$ – a learnable attention matrix $\implies$ graph attention network

Implicit and multi-step explicit schemes does not have that property!

The GRAND Framework

Main Idea

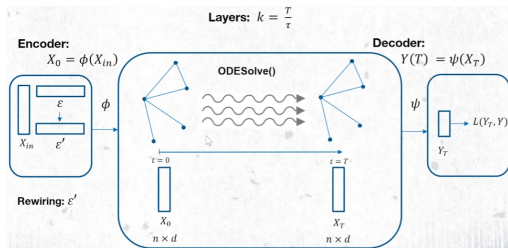
- Model deep learning on graphs as a **continuous diffusion process**, and treat GNNs as its discretisation.
- This allows leveraging the vast and mature field of numerical analysis to build better GNNs.

Architecture

1. **Encoder ϕ :**
 $X(0) = \phi(X_{in})$.
2. **ODE Solver:**
Integrates the dynamics from $t = 0$ to a terminal time T .

$$X(T) = X(0) + \int_0^T \frac{dX(t)}{dt} dt$$

3. **Decoder ψ :**
 $Y = \psi(X(T))$.



Spatial Discretisation: Graph Rewiring

Decoupling Computation from Input

- ▶ The diffusion process can happen on a different computational graph $G' = (V, E')$ instead of the input graph $G = (V, E)$.
- ▶ Analogous to choosing a different spatial discretisation grid in PDEs

Why Rewire?

- ▶ Denoise the input graph structure.
- ▶ Avoid information bottlenecks.¹
- ▶ Improve numerical efficiency.

GRAND-nl-rw Implementation

A two-step rewiring process:

1. Densify the graph based on feature similarity (e.g., k-NN).
2. At runtime, use a subset of edges where attention $a_{ij} > p$.

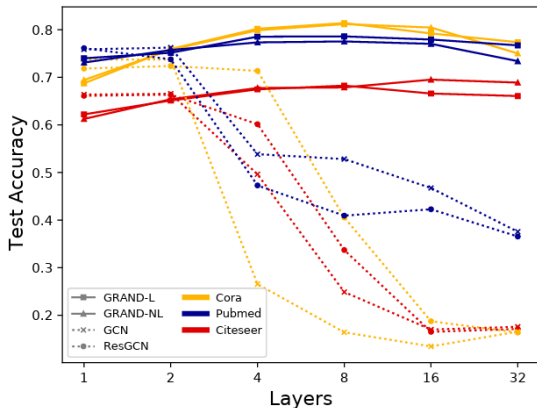
$$\frac{dx_i(t)}{dt} = \sum_{j:(i,j) \in E', a_{ij} > p} a_{ij}(t)(x_j(t) - x_i(t))$$

¹Understanding over-squashing and bottlenecks on graphs via curvature

Experiments & Results

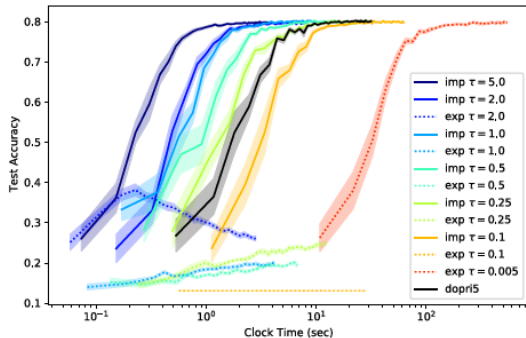
Solving Oversmoothing with Depth

- ▶ GRAND maintains performance at large integration times T (equivalent to deep networks).
- ▶ GCN/ResGCN performance degrades sharply after few layers.



Experiments & Results

Solver Choice and Performance



- **Explicit:** Fast but only stable for small τ .
- **Implicit:** Stable for all τ , can be faster for large steps.
- **Adaptive:** Robust default.

Side note:

Beltrami Flow and Neural Diffusion on Graphs²

Core Idea

- ▶ Treat the graph as a discrete sample of an underlying continuous manifold.
- ▶ Add positional embedding that represents graph structure
- ▶ Govern feature and graph structure evolution jointly by the *Beltrami flow*

The Beltrami Flow Equation

$z_i = (u_i, \alpha x_i)$, $(x)_i$ – node features, $(u)_i$ – positional features.

$$\frac{\partial z_i(t)}{\partial t} = \text{div}(a(z(t)) \nabla z_i(t))$$

$$\left(\frac{\partial U(t)}{\partial t}, \frac{\partial X(t)}{\partial t} \right) = (A(U(t), X(t)) - I) (U(t), X(t))$$

$$U(0) = U, \quad X(0) = \alpha X, \quad t \geq 0$$

²<https://arxiv.org/pdf/2110.09443>